

Ana-Catalina Villalobos Contardo

DATA SCIENTIST & MACHINE LEARNING ENGINEER

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SUMMARY

Civil Engineer with a solid background in Data Science, Machine Learning (ML), Data Engineering, and Backend Development. Combines a strong theoretical foundation (mathematics, statistics) with practical experience in developing and implementing ML solutions (predictive maintenance, operational optimization). Specialist in designing robust data pipelines (ETL/ELT), interpretable predictive models, and complex integration architectures. Proven technical leadership managing multidisciplinary teams, with a clear focus on solving complex problems, generating strategic value, and operational excellence.

EXPERIENCE

SimpliRoute | *Learning Engineer* | August 2025 - Present

- Architecture and implementation of a fault-tolerant, multi-provider geolocation and geocoding platform, designing decoupled asynchronous pipelines, high-availability fallback strategies, and standardized microservices.
- Design and production deployment of enterprise agentic infrastructure leveraging Model Context Protocol (MCP) for logistics engines, establishing deterministic guardrails, least-privilege tool execution boundaries, and distributed state persistence on Kubernetes with Redis.
- Large-scale data modeling and event orchestration in Google Cloud (BigQuery, Pub/Sub, and Apache Airflow), ensuring data lineage, operational reliability, and real-time telemetry for optimization systems.
- Leadership in Generative AI adoption across development lifecycles (DevEx) and core systems, integrating agentic workflows to streamline automated testing, accelerate delivery cycles, and enhance operational efficiency.

Fractal | *Tech Lead Fractal Hub* | November 2023 - July 2025

- Led a multidisciplinary integration team, achieving product development goals and client projects, while prioritizing team mentoring.
- Strategic planning of the product roadmap, ensuring alignment with business objectives and continuous value delivery.
- Python development for data processing and orchestration libraries, enabling integrations with multiple systems.
- Supervision of integration projects, ensuring deadlines and quality in data flows across heterogeneous systems.

Fractal | *Data Scientist* | August 2021 - November 2023

- Designed and implemented data science processes for predictive maintenance, asset prioritization, and failure estimation.
- Collaborated on product creation and development to manage integrations between Fractal and other platforms.

Fractal | *Data Analyst* | February 2020 - July 2021

- Developed a predictive maintenance tool using advanced data analysis, predictive statistics, and Machine Learning techniques.
- Database design and management alongside web development tasks.

TOOLS & TECHNICAL SKILLS

- **Programming Languages:** Python (Principal/Advanced) | SQL | JavaScript | TypeScript | MATLAB | Java | C# (.NET) | Rust.
- **Cloud & DevOps:** Google Cloud Platform (BigQuery, Cloud Run, Artifact Registry, Google Pub/Sub, Cloud Storage, GKE) | ArgoCD | Git | GitHub | Bitbucket.
- **Frameworks & Libraries:** FastAPI | Pydantic | Pandas | NumPy | Scikit-learn | .NET 9 (WPF / WinUI 3) | FastHTML | LangChain | TensorFlow | Django | React JS.
- **Databases:** PostgreSQL | Redis | BigQuery | SQL Server | MySQL | MongoDB | SQLite.
- **Containers & Orchestration:** Airflow | Airbyte | Docker | Kubernetes.
- **Management & Communication:** Jira | MS Planner | MS Teams | Sharepoint | Notion | Slack | Trello.
- **Methodologies:** Test-Driven Development (TDD) | Agile Methodologies (Scrum & Kanban) | Roadmap Planning.
- **Key Skills:** Model Context Protocol (MCP) | Multi-Agent Systems & LLMs | ML Model Monitoring | Data Extraction & Processing | Data Visualization | Relational DB Design | Technical Leadership | Agentic AI.

EDUCATION

- **Universidad de Chile** | *Civil Engineering with a minor in structures, construction, and geotechnics* | Graduated 2019, Degree awarded 2020 (Maximum Distinction)
Thesis: Gaussian Processes and vibration modes for delamination detection in composite material plates.
- **Universidad de Chile** | *Bachelor of Engineering Sciences, Civil minor* | Completed 2017
Notable activity: Teaching Assistant for Probabilities and Statistics.

PUBLICATIONS

- **Generalized Gaussian smoothing for baseline-free debonding assessment of sandwich panels** | *Structural Control Health Monitoring* - March 2021 (<https://doi.org/10.1002/stc.2727>)
Methodological proposal for delamination detection and evaluation in composite plates using Gaussian Processes and Bayesian analysis.

LANGUAGES & CERTIFICATIONS

- **Languages:** Spanish (Native) | English (Advanced - EFSET English Certificate C2 Proficient, 2020)
- **Certifications:** React & Django Full Stack (Udemy, 2022) | Mastering Data Visualization (Udemy, 2022) | Applied Machine Learning in Python (Coursera, 2020).